

```

1 #!/usr/bin/python
2 import os
3 import json
4 import sys
5 sys.path.insert(0, '../bluetooth')
6 import bluetooth_gap
7 import bluetooth_constants
8
9 if 'REQUEST_METHOD' in os.environ:
10     result = {}
11     args = json.load(sys.stdin)
12     if os.environ['REQUEST_METHOD'] != 'PUT':
13         print 'Status: 405 Method Not Allowed'
14         print
15         print "Status-Line: HTTP/1.0 405 Method Not Allowed"
16         print
17     else:
18         print "Content-Type: application/json;charset=utf-8"
19         print
20         if not "bdaddr" in args:
21             result['result'] = bluetooth_constants.RESULT_ERR_BAD_ARGS
22             print json.JSONEncoder().encode(result)
23         else:
24             bdaddr = args["bdaddr"]
25             rc = bluetooth_gap.connect(bdaddr)
26             result['result'] = rc
27             print json.JSONEncoder().encode(result)
28     else:
29         print "ERROR: Not called by HTTP"

```

do_connect.py

Figure 9: Adapter code

```

1 #!/usr/bin/python
2 import bluetooth_constants
3 import bluetooth_general
4 import bluetooth_exceptions
5 import bluetooth_utils
6 import sys
7 import dbus
8 import dbus.mainloop.glib
9 from dbus import ByteArray
10
136 def connect(bdaddr):
137     bus = dbus.SystemBus()
138     device_proxy = bluetooth_general.getDeviceProxy(bus, bdaddr)
139     if device_proxy == None:
140         return bluetooth_constants.RESULT_ERR_NOT_FOUND
141     device_path = device_proxy.object_path
142     if not bluetooth_general.is_connected(bus, device_path):
143         try:
144             device_proxy.Connect()
145         except Exception as e:
146             return bluetooth_constants.RESULT_EXCEPTION
147         return bluetooth_constants.RESULT_OK
148     else:
149         return bluetooth_constants.RESULT_OK
150

```

Bluetooth API connect function

Figure 10: API code

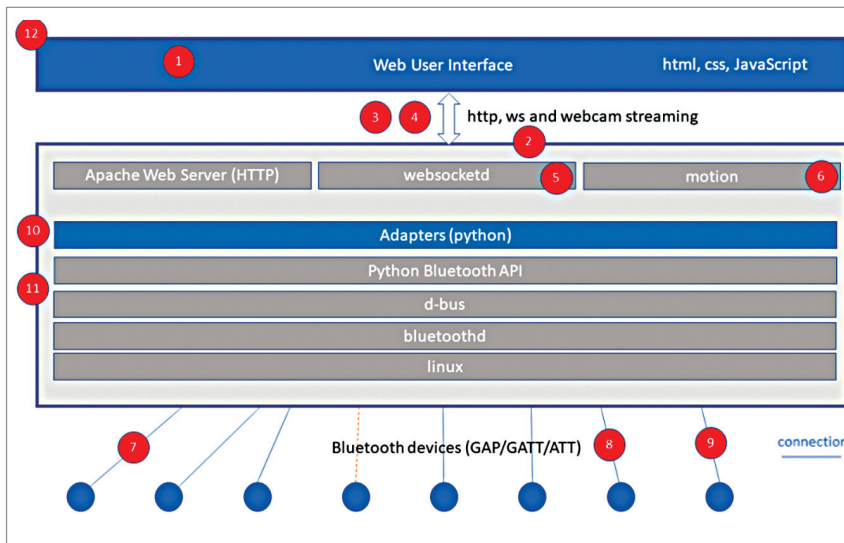


Figure 11: Security analysis

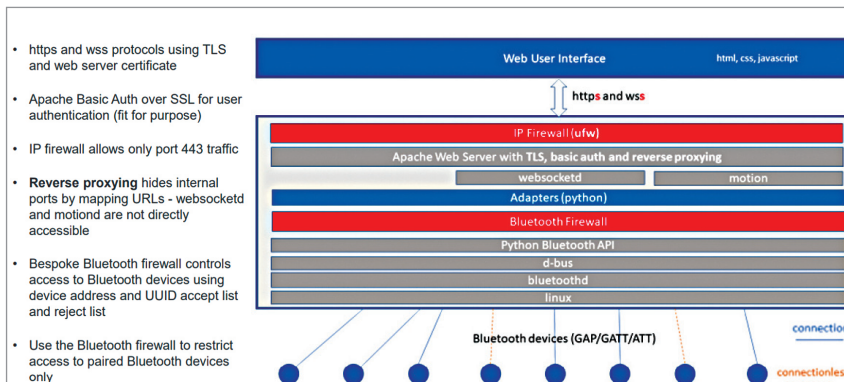


Figure 12: Steps for securing the gateway

In this example, HTTP PUT was used (Figure 8) for writing to a Bluetooth characteristic. Here the Bluetooth device address is assigned as the first parameter and identifies the device to interact with.

The handle parameter is a unique identifier for a specific instance of a Bluetooth characteristic, which belongs to a particular Bluetooth service.

Bluetooth Internet gateway security

Anything that allows access to your device over the Internet needs to be secure. A review of the security (Figure 11) of the initial gateway design and implementation identified numerous security issues.

For example, there was no authentication mechanism and therefore no control over who may or may not access the gateway.

The gateway at this stage used three different TCP/IP ports: port 80 for HTTP, 8081 for the webcam streaming service, and 8082 for websocketd. The more ports exposed directly to the Internet, the greater is the security risk.

Data over TCP/IP is not encrypted at this stage either and nor is data over Bluetooth necessarily.